Pranjal Patel

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Education

Rowan University Glassboro, NJ
Master of Science in Computer Science *May 2025*

Gujarat Technological University Ahmedabad, India
Bachelor of Engineering in Computer Engineering *May 2022*

Technical Skills

Programming & Data: Python, R, SQL, Pandas, NumPy, Dask, PySpark

Machine Learning & AI: Scikit-learn, PyTorch, TensorFlow, OpenCV, XGBoost, LightGBM, Gradient Boosting, Time-Series Forecasting, Anomaly Detection, Image Segmentation, Transformers, Hugging Face, LLMs, RAG, LangChain, LlamaIndex, Prompt Engineering, FAISS, Vector Databases

Big Data & Cloud: Apache Spark, Kafka, Airflow, Databricks, AWS (S3, EC2, SageMaker, Lambda, Step Functions)

MLOps & Data Engineering: Docker, Kubernetes, MLflow, FastAPI, CI/CD, Feature Stores, Snowflake, Redshift, ETL/ELT, Weights & Biases

Experience

Machine Learning Engineer July 2025 – Present
BlackRock *NY*

- Designed and deployed production fraud detection models across transaction monitoring systems to reduce false positives by 30% and lower AML manual review load.
- Automated end-to-end ML pipelines with CI/CD integration to eliminate multi-day manual model release cycles and accelerate deployment across teams.
- Engineered real-time anomaly detection for high-volume transaction streams to deliver sub-second inference and strengthen compliance and risk monitoring.
- Built a RAG-based document search system using LlamaIndex and FAISS on AWS to reduce research turnaround by 40% and enable semantic retrieval across internal knowledge bases.

Machine Learning Engineer May 2022 – Aug 2023
Cred *India*

- Fine tuned LLM-based NLP classifiers with Hugging Face across multilingual datasets to improve prediction accuracy by 22% and replace legacy rule-based logic.
- Built distributed ETL pipelines with Apache Spark and Airflow to streamline data processing and support reliable, high-volume model training across teams.
- Designed recommendation engines with collaborative filtering and LightGBM to strengthen personalization quality and lift engagement on customer-facing platforms.

Machine Learning Engineer Intern July 2020 – May 2021
Cred *India*

- Researched and built YOLO-based detection models for driver monitoring in self-driving systems to track face detection, gaze, eye closure and posture and flag driver drowsiness and distraction for a Y Combinator-backed startup.
- Optimized and deployed the detection pipeline across cloud and edge computing devices to deliver low-latency inference and own the path from research prototype to production system.

Projects

Morph & Split – End-to-End ML Dataset Preparation Tool | *Python, TensorFlow, LLM Integration, Full-Stack Web*

- Developed a full-stack web application to streamline image-mask preprocessing for computer vision workflows and accelerate ML data preparation.
- Implemented stratified dataset splitting, augmentation, cropping, and resizing with automatic TensorFlow-ready output to ensure reproducibility and error-free pipelines.

Brain Tumor Segmentation from MRI Scans | *Python, PyTorch, nnU-Net, Medical Imaging*

- Built an nnU-Net segmentation model to automatically delineate brain tumors from T1, T2, and FLAIR MRI scans and reduce reliance on manual radiologist annotation.
- Trained and evaluated the model on the BraTS dataset to segment tumor core, edema, and enhancing regions with strong overlap accuracy.